Whose Regulation Does the Model Know? Reliability of LLM-Retrieved Regulatory Information for Public Environmental Management in 25 Countries

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Abstract

Public officials increasingly consult large language models (LLMs) about the regulations they are responsible for applying, and generative AI is already in routine administrative use [1]. When the answer is wrong, the official carries the consequence [2], and technology-neutral regulation does not reach the failure [3], which leaves verification at the point of use as the operative control. Whether that reliability is distributed evenly across the states that depend on it is unmeasured, because existing evaluations score perception, explanation quality or generic factual recall rather than agreement with the binding text.

We test this where the correct answer is published in an official register and can be checked: environmental regulation. No prior benchmark combines a regulated policy domain, official-source ground truth, a pre-specified persona factor, and a mechanism test that separates language-corpus size from country coverage. The design spans **25 countries, 14 LLMs, four languages, and five task types** (normative recall, local measured data, health-evidence synthesis, policy instruments, and applied recommendation). We evaluate each model *as served* through a fixed access stack (vendor API or pinned inference endpoint), with a within-subject persona factor (neutral versus public environmental manager) and a within-country English/native-language contrast. The design was pre-specified for 15 countries and then extended post hoc to 25 (seven more Global North, three more Global South) to gain power; every effect is reported alongside its pre-specified 15-country value. Ground

truth for the two numeric tasks is adjudicated against official registers by code rather than by a language model, and the residual is scored by the mean of a three-vendor judge panel.

The study's most robust finding is that **the fixes practitioners reach for do not work, and the most intuitive one backfires**. Asking in the country's own language **lowers** accuracy by 5.0 percentage points (Wilcoxon signed-rank $p < 10^{-5}$, $n = 839$ matched model-prompt cells), and the penalty is larger the scarcer the language's corpus: Hindi -12.0 pp, Portuguese -3.9 pp, Spanish -4.6 pp, all significant. A local-manager persona does not narrow the gap either (permutation $p = 0.22$), and the Brazilian-Portuguese regional model is the weakest of all fourteen (Cliff's $\delta = -0.45$). All three remedies an agency would try — speak the local language, declare the official role, procure a regionally tuned model — fail together, and the first actively harms.

Two further harms are robust. **Accuracy collapses on the recall of a binding value**: technical-standard recall scores 0.37 against 0.62 for synthesis and recommendation (Cliff's $\delta = -0.47$), a regulatory-cutoff floor exactly where an administration needs the model most. And a Global North/South **tier** gap of +4.6 pp persists (bootstrap 95% CI $[+1.2, +7.8]$, excludes zero), modest on the composite and specification-sensitive there, but concentrated rather than weak: estimated task by task, the Global South odds of returning the binding national standard are 0.23 of the Global North odds ($[0.19, 0.28]$), and the gap shrinks as the answer depends less on a country-specific fact, vanishing on the one task with no register value to get right.

What that tier gap is *made of* we report as exploratory. A monotonic development gradient is suggestive but does not reach significance (Spearman $\rho = 0.36$ with HDI, $p = 0.080$; Mann-Kendall $p = 0.072$; at the pre-specified $n = 15$, $\rho = 0.08$), so we describe a tier difference we can demonstrate and a gradient we cannot. The mechanism is likewise unresolved: the pre-specified language-corpus proxy is null ($\rho = 0.09$), and a country-coverage proxy is no longer significant either once the ground truth is corrected ($\rho = 0.35$, $p = 0.087$; partial $\rho = 0.17$, $p = 0.44$ after adjusting for development). The design also cannot separate the linguistic channel at country level, because the Joshi variable takes three values across 25 countries, eighteen of them the same.

Reliability is reported on the sample that produces the effects rather than on a calibration subset: across 3,190 items scored by all three judges, panel-mean ICC$(2, 3) = 0.79$ and single-judge ICC$(2, 1) = 0.56$, which is why the analysis uses the panel mean and not one judge. Ground truth is anchored to official primary sources for all 25 countries. Beyond its findings, the study offers a reusable, fully reproducible template — official-source ground truth, code-adjudicated scoring where the answer is a number, multi-method robustness, and an end-to-end audit pipeline — for applied LLM-equity evaluation. Protocol, prompts, and analysis code will be released on publication.

Keywords: generative AI in government; regulatory information; information reliability; public environmental management; algorithmic decision support; large language models; Global South

1 Introduction

Large language models (LLMs) have moved from research curiosity to working infrastructure of public administration in under five years: an OECD survey of core government functions documents generative-AI adoption across policymaking and public-service delivery, including deployments that help public-administration staff search regulations and draft technical summaries [4].

Environmental agencies, municipal secretariats, and applied policy analysts increasingly draft technical syntheses, translate regulatory norms, locate official monitoring data, and characterize institutional frameworks with such assistance. Air pollution policy is a domain where this adoption carries direct stakes for public health. Fine particulate matter ($PM_{2.5}$) is the leading environmental risk factor for premature mortality worldwide [5, 6], and the burden falls disproportionately on the Global South [5]. When a public environmental manager in São Paulo, Lagos, or Jakarta asks an LLM about an ambient air quality standard, an attainment deadline, or the epidemiological evidence behind an intervention, an inaccurate answer is not a benign error. It can propagate into a regulatory brief, a procurement decision, or an emergency response during a pollution episode.

The question this paper addresses is whether LLMs serve Global South air pollution policy as reliably as they serve the Global North, and whether the way a query is framed changes the answer. Two recent strands of evidence make this question urgent rather than hypothetical.

Geographic factual bias is real and measurable. Moayeri et al. [7] documented 1.5× higher factual error rates for Sub-Saharan African countries versus North American countries across 20 models and 11 World Bank indicators. Manvi et al. [8] showed that LLM zero-shot ratings on subjective topics correlate strongly with income level (ρ up to 0.70 against socioeconomic conditions). Mirza et al. [9] found that GPT-4 systematically privileges statements about the Global North over the Global South in factual verification tasks. These results establish the phenomenon, but none of them targets a single high-stakes applied domain with verifiable official ground truth, and none asks whether the bias can be modulated by prompt framing.

Linguistic and corpus asymmetry is structural. Joshi et al. [10] mapped a six-class hierarchy of linguistic resource availability in NLP, showing that the languages of most Global South countries sit in the under-served lower classes. Under-representation in the training corpus is the dominant explanation for downstream geographic bias [11]. This asymmetry is not incidental; it reproduces longstanding patterns of uneven digital and epistemic infrastructure that decolonial scholarship has analysed for AI systems [12].

We identify three gaps that current work leaves open for a domain like air pollution policy.

Gap 1: no benchmark targets air pollution policy with official ground truth. Existing evaluations use numeric World Bank recall (WorldBench), subjective ratings (Manvi et al.), journalistic fact-checking (Global-Liar), or everyday cultural knowledge (BLEnD [13]). None measures the recall and synthesis tasks a public environmental manager actually delegates to an LLM: stating a binding ambient standard, retrieving an officially measured concentration, summarizing the health-burden evidence, identifying the legal instruments and executing agencies, or recommending an operational response. These tasks anchor to authoritative sources (national environmental councils, state monitoring agencies, the World Health Organization, the Global Burden of Disease estimates), which makes accuracy verifiable in a way that trivia benchmarks are not.

Gap 2: persona framing is unexamined as an experimental lever. Practitioners routinely prefix queries with a role ("As a municipal environmental secretary, ..."), and persona or role conditioning is a documented prompt pattern in deployed prompting workflows [14]. Whether adopting the persona of a public environmental manager *reduces* the Global South accuracy gap (by steering the model toward policy-relevant, regulation-compliant responses) or *amplifies* it (by inviting confident fabrication of regulatory specifics the model never learned) has not been tested in a pre-specified design. We treat persona as a within-subject experimental factor rather than an uncontrolled confound.

Gap 3: the corpus-representation mechanism is asserted, not tested for this domain. The link between corpus representation and accuracy is assumed but rarely measured against confounders, and rarely separated into its two channels. We distinguish *language-corpus* size (how much text exists in a country's language: mC4 token volume [15], OSCAR size [16], Joshi class) from *country-coverage* breadth (how broadly the corpus is about a country: Wikipedia and Wikidata coverage), and test which predicts air pollution policy accuracy after adjusting for Human Development Index — a distinction that matters because the geographic gap is measured in English.

Contributions of this work

We present a benchmark designed for these gaps, with hypotheses, sampling, and analysis plan pre-specified before data collection [17], executed across 15 countries and then extended to 25. Throughout, our unit of analysis is the model *as served* — a stack of model weights, vendor, and serving infrastructure reached through a pinned API or inference endpoint — rather than an idealized set of weights, because a public manager interacts with the served stack and any measured gap is a property of that stack (Section 3). Our contributions are:

1. **An auditing protocol for regulatory information, not a trivia benchmark.** The reliability question an administration faces is not whether a model writes well about policy but whether the value it returns matches the one published in the official register. We build five task types from the everyday work of an environmental agency: the binding ambient standard in force (T1), the officially measured local concentration (T2), the health-burden evidence an impact assessment cites (T3), the national policy instruments and their executing agencies (T4), and an applied recommendation for an operational scenario (T5). Each key is read from an official source, so correctness is adjudicable rather than a matter of expert impression. The protocol transfers to any regulated domain where the correct answer is published: tax, licensing, public health, procurement.
2. **The three localisation fixes agencies reach for, tested rather than assumed — and all three fail.** Practitioners respond to poor answers about their own country with three moves: tell the system you are the responsible official, ask in the national language, or adopt a regionally tuned model. Each is a procurement or workflow decision with a cost, and each circulates as received wisdom. We test all three: persona as a within-subject experimental factor, native-language prompting against the English baseline within

country, and a Brazilian-Portuguese instruction-tuned model against a scale-matched global one. None works, and the one a manager is most likely to try first does active harm: asking in the country's own language lowers accuracy by five percentage points, most sharply for the language with the scarcest corpus. This is the study's principal finding and its most directly actionable output for practice.

3. **Model choice as an administrative decision.** Whether an agency runs an open-weight model on its own infrastructure or a vendor-hosted one is now an explicit procurement question [18]. We evaluate fourteen deployment stacks spanning that spectrum, and treat the unit of analysis as the model *as served*, since an agency contracts a served stack and not a set of weights.

4. **A sampling design that separates who is poorly served from why.** Countries are stratified along independent axes: UNCTAD North/South classification (operationalising coloniality [12]), the Joshi et al. linguistic-resource taxonomy [10], and development indicators. This lets geographic, linguistic and economic explanations be told apart instead of conflated, which matters for policy: each implies a different remedy and a different responsible actor.

5. **A mechanism test that reports its own limits.** We test corpus representation using both language-corpus and country-coverage proxies, with partial-correlation controls. We report where the pre-specified proxy fails, where an alternative is merely suggestive, and where the instrument itself lacks the resolution to decide, rather than presenting the surviving correlation as an established cause. On this evidence the country-level mechanism is unresolved, and we separate throughout what the design demonstrates — a tier difference, a language penalty, a recall floor — from what it cannot, namely the shape of the gradient across the development range and the corpus channel behind it.

6. **Ground truth adjudicated by code, and what that changes.** Where a task's correct answer is a number published in an official register, the verdict is computed against that register rather than delegated to a language model, because comparing a value with an authorised range is exact and a judge can only reproduce it with noise. This is not a detail of implementation. On the same responses, rebuilding two task keys from official registers and moving the comparison into code cut the development gradient from $\rho = 0.51$ to $\rho = 0.36$ and more than doubled the native-language penalty. In this literature the ground-truth key, not the sample size, is the binding constraint.

7. **Open resources.** The full protocol, prompts, ground-truth registries with their official sources, analysis code, and the deviation log will be released under CC-BY-4.0 and MIT licenses.

Theoretical framing

We interpret our findings within the *coloniality of knowledge* framework [12, 19] and its data and infrastructural articulations [20, 21]. LLMs are compressed statistical summaries of corpora produced under conditions of uneven digital infrastructure and uneven publication access. The languages and regions that Joshi et al. place in the under-served classes are precisely those where air pollution kills the most people [5, 10]. When a Global South environmental manager consults a model trained largely on Global North text to govern a local air quality problem, the resulting decision risks inheriting the asymmetries of the corpus. This positions our empirical work as a test of one link in a theorised reproduction cycle (training-corpus asymmetry → LLM knowledge asymmetry → policy decision asymmetry), and the persona manipulation as a test of whether prompt framing can interrupt that link.

The paper proceeds as follows. Section 2 positions our work relative to the literature. Section 3 presents the pre-specified design. Section 4 reports the empirical findings. Section 5 interprets them through both statistical and critical lenses and considers practical implications for air pollution governance. Section 6 closes.

2 Related Work

We organize prior work along four threads: the discovery and measurement of geographic factual bias, multilingual and cultural extensions, alignment and mitigation attempts, and LLM use in policy and health decision contexts. For each we state how the air pollution policy focus and the persona manipulation of the present study extend the thread.

2.1 Geographic factual bias

Manvi et al. [8] established that frontier LLMs carry systematic biases against regions of lower socioeconomic conditions, with Spearman correlations up to 0.70 between model-generated ratings of attractiveness, morality, and intelligence and a proxy for those conditions. The outcome space (subjective ratings) leaves open whether the same bias appears in objective recall of regulatory and epidemiological facts, which is what air pollution policy requires.

Moayeri et al. [7] introduced WorldBench, quantifying factual recall against World Bank indicators across 20 LLMs and 11 indicators and documenting 1.5× higher absolute relative error for Sub-Saharan Africa versus North America. WorldBench also enables automatic detection of citation hallucination, where models cite the World Bank while producing false statistics. The design is bounded by the macro-economic indicator space; an environmental manager asks narrower, source-specific questions (a national ambient standard, an attainment phase deadline, a city-year measured concentration) whose ground truth lives in environmental councils and monitoring agencies rather than in a single aggregated database.

Mirza et al. [9] (Global-Liar) extended the question to temporal stability, showing that newer GPT-4 versions do not monotonically improve and that the privileging of Global North statements persists across versions. Their contribution is temporal and within-model rather than cross-model, cross-language, or domain-specific.

These studies establish that the phenomenon exists and is measurable. None isolates a single applied domain with verifiable official ground truth, and none treats prompt framing as a manipulable factor.

2.2 Multilingual and cultural extensions

Myung et al. [13] (BLEnD) introduced question-answer pairs covering 16 countries and 13 languages in everyday cultural knowledge. Their finding that LLMs perform better in the native languages of high-resource cultures but worse in those of low-resource ones motivates our exploratory language hypothesis (H2) and connects directly to the resource hierarchy of Joshi et al. [10]. Yet BLEnD measures cultural-knowledge accuracy, not the accuracy of regulatory and health-policy recall, and its country selection is not stratified theoretically.

Regional benchmarks proliferated: BRoverbs [22] for Brazilian proverbs, TiEBe [23] for regional historical events, and ALBA [24] and AMALIA [25] for European Portuguese. These document concrete failures in culturally grounded tasks but do not address applied policy use, and each is limited to one or two languages. Work on Portuguese corpus construction [26] underscores how Common Crawl snapshots under-represent Lusophone content, the corpus asymmetry our H4 tests.

2.3 Alignment and mitigation

Opuszko and Böhm [27] investigated whether reasoning models reduce geographic bias, finding that reasoning helps in aggregate but does not eliminate regional disparities. Kerche et al. [28] proposed a place-based typology of LLM biases, situating geographic bias within a broader sociotechnical landscape. These contributions are conceptual and aggregate; we add a domain-specific, pre-specified empirical test and introduce persona as a candidate mitigation lever rather than a fixed property of the model.

2.4 LLMs in policy and health decision contexts

A parallel literature examines LLM use in specific policy settings. Le Coz et al. [29] evaluated LLM policy recommendations for homelessness across four cities (including Barcelona, Johannesburg, and South Bend), finding "context-blind rigidity" in which models apply stable internal heuristics poorly calibrated to local realities and align more closely with Global North experts. Their result shows the bias matters for policy in a single domain and a single task type (recommendation). We generalize to a regulatory and health domain with five task types spanning recall, synthesis, and recommendation, and we add the persona manipulation absent from their design.

In health specifically, Kozlakidis et al. [30] argue that LLM hallucinations carry acute risk in regulatory environments with limited technical capacity, noting that language and cultural mismatch between globally trained models and local practice increases error. Air pollution policy sits at the intersection of this risk and the mortality burden documented by the Global Burden of Disease estimates [5], which makes accurate retrieval of standards and health evidence a public-health concern rather than a benchmarking abstraction.

2.5 Persona and role conditioning

Role or persona conditioning is a documented prompt pattern in deployed prompting [14], and a growing body of work reports that assigning an expert role can shift model outputs in either direction depending on task and domain. Systematic evidence is sobering: across four LLM families and 2,410 factual questions, adding expert personas to system prompts did not improve accuracy over a no-persona control, and across 162 roles it slightly *reduced* accuracy on average [31]; an independent replication likewise finds that expert personas do not improve factual accuracy [32]. Whether persona conditioning helps or harms factual reliability in high-stakes, knowledge-intensive domains is therefore unsettled and, to our knowledge, untested for geographic equity. We therefore treat persona as a pre-specified experimental factor and test both the Norm-Compliance hypothesis (persona reduces the gap) and the Persona-Vacuum hypothesis (persona amplifies it).

2.6 Generative AI inside the administration

The deployment question is no longer hypothetical. Kim [1] estimates bureaucratic productivity effects of generative AI in public administration quasi-experimentally, and Robinson [18] documents that choosing which model an agency runs is now an explicit procurement decision, weighed between open-weight and vendor-hosted options. Both of our model-level findings speak directly to that decision: the openly available frontier models trail the closed ones, and the regionally tuned Portuguese model is the weakest of the fourteen we evaluate, so the two choices an agency might make on sovereignty or locality grounds are the two that cost the most accuracy.

Two further strands frame why a wrong answer matters administratively rather than merely technically. Choroszewicz [2] shows that early generative-AI support tools position frontline practitioners as "moral crumple zones", absorbing responsibility for failures originating upstream; a civil servant who files a brief citing a superseded emission limit occupies exactly that position. And Cordella and Gualdi [3], analysing Italy's intervention on ChatGPT, argue that technology-neutral regulatory frameworks do not reach the specific harms these systems produce, which leaves verification at the point of use as the operative control. The literature on how such decisions are received is correspondingly developed: Aoki et al. [33] test whether the type of explanation changes perceived accuracy, fairness, and trustworthiness of an algorithmic decision among those affected by it.

What none of this work establishes is whether the underlying answer is *correct*, and whether correctness is distributed evenly across the states that would rely on it. Perceived trustworthiness and measured accuracy are different quantities, and a system can be well explained, well governed, well received, and wrong. That is the gap this study addresses, using a domain where the correct answer is published in an official register and can be checked.

2.7 Positioning of the present work

Our design differs from prior work in four concrete ways: (i) a single high-stakes applied domain (air pollution policy) with verifiable official ground truth, rather than numeric trivia or cultural knowledge; (ii) persona as a pre-specified within-subject experimental factor, rather than an uncontrolled framing choice; (iii) theory-driven country sampling on independent axes (UNCTAD coloniality, Joshi linguistic resource, development), rather than regional convenience; and (iv) an explicit corpus-representation mechanism test with sensitivity analysis. No prior study combines a regulated public-health domain, a persona manipulation, and a pre-specified mechanism test.

3 Methods

3.1 Research design

We specify a pre-specified, factorial, multi-country benchmark experiment to quantify geographic and persona-modulated bias in 14 large language models on air pollution policy tasks relevant to public environmental management. The pre-specified design crosses 15 countries (stratified along three theoretically grounded axes), 14 LLMs spanning five access tiers, one theory-driven domain (air pollution policy), 5 task types, and 2 persona conditions (neutral versus public environmental manager), with 2 stochastic replications per prompt-model-persona cell; a post-hoc extension (Section 3.3) enlarges the sample to 25 countries. The stimulus set crosses each country's five tasks and two personas in English, plus a within-country native-language rendering for the nine countries with a target native language, scored on the $[0, 1]$ composite. The study protocol fixes, in advance of data collection, the hypotheses, primary models, persona protocol, exclusion criteria, and multiple-comparison corrections. A Brazil-only collection executed under an earlier three-domain design is reclassified as an extended pilot and is reported separately; it informed prompt calibration but contributes no confirmatory observations.

3.2 Country selection

Countries were selected via theoretical sampling [34] triangulated across three independent classifications. First, the **UNCTAD Global North/South classification** [35] operationalizes the coloniality-of-knowledge framework that motivates this study [12]. Second, the **Joshi et al. (2020) six-class taxonomy** of linguistic-resource representation in NLP corpora provided an orthogonal stratification dimension [10]. Third, **World Bank income groups (FY26)** [36] captured economic position independently of the previous two axes.

The final sample of 15 countries — Brazil, Mexico, Argentina, Peru (Latin America); Nigeria, South Africa, Kenya, Egypt (Africa); India, Indonesia, Bangladesh, the Philippines (Asia); and the United States, Germany, Japan (Global North reference) — covers four world regions and four World Bank income groups. The linguistic-resource axis is far less differentiated than the sampling frame intended: the dominant official languages of these countries fall into only three Joshi classes, and twelve of the fifteen share Class 5. Extending to 25 countries does not help, since eighteen of the twenty-five are Class 5. This is a property of the world, not of our draw, because the official languages of most states are themselves high-resource; but it means the linguistic axis has too little variance to be separated from the others, and we treat the Joshi results accordingly (Section 4.6). The twelve Global South countries also vary in domestic air quality governance maturity, which is relevant to ground-truth availability (Section 3.9). Countries with compromised statistical governance (e.g., active conflict) or without a publicly retrievable national ambient air quality standard were excluded; Oceania and Joshi Class 0 languages were not represented given feasibility constraints and are acknowledged as scope limitations.

3.3 Confirmatory sample expansion (post-registration)

After the pre-specified 15-country analysis, and *transparently reported here as a post-hoc extension rather than part of the locked protocol*, the sample was enlarged to **25 countries** to increase statistical power for the gradient (H1) and corpus-mechanism (H4) tests and to multiply the within-country language pairs available for H2. Ten countries were added under the identical prompt-construction, ground-truth, collection, and scoring procedures. Six are **Global North** references chosen purely to broaden the high-resource end of the gradient — the United Kingdom, Canada, Australia, South Korea, France, and Italy. Four are **native-language-pair** countries that add Spanish (Colombia, Chile) and Portuguese (Portugal, Angola) contrasts for H2; of these, Portugal is also Global North while Colombia, Chile, and Angola are Global South. The two roles overlap, so by tier the extension adds seven Global North countries (the six references plus Portugal), raising the Global North cell from three to ten, and three Global South, while by language it raises Spanish-speaking pairs from three to five and Portuguese-speaking pairs from one to three. Because this extension was decided after observing the initial results, every effect computed on the 25-country sample is reported alongside its pre-specified 15-country counterpart, and no pre-specified conclusion is revised on the basis of the extension alone; the expansion is treated as a power-and-robustness check, not as a new confirmatory test. Angola, like Nigeria and Argentina, has *no* national ambient $PM_{2.5}$ standard, providing additional no-standard cases in which a faithful model must decline to state a non-existent threshold rather than fabricate one.

To multiply H2 pairs without adding new languages, two additional question variants per (country, task) were also generated for the five original native-pair countries via a high-quality generator model and rendered in both English and the native language; these variants are used *only* for the paired English-versus-native contrast (H2), in which any imperfection in the generated ground truth cancels between the two members of a pair, and are excluded from absolute-accuracy comparisons (H1, H3).

3.4 Country covariates

For each country we compiled developmental and corpus-representation covariates. **Human Development Index (HDI)** (UNDP Human Development Report 2023–24, 2022 data) and **GDP per capita** index developmental position and enter the mechanism analysis as adjustment covariates.

The corpus-representation exposures are organized along a distinction that the mechanism analysis (H4) makes explicit, because the two channels apply to different effects. **Language-corpus measures** quantify how much pretraining text exists *in* a language and are the candidate mechanism for the native-language penalty (H2): the mC4 token count per language (15, Table 6), the OSCAR-2301 deduplicated corpus size per language [16], and the Joshi linguistic-resource class [10]. **Country-corpus measures** quantify how much the encyclopedic corpus is *about* a country, independent of language, and are the candidate mechanism for the geographic gap (H1/H4): the byte length of the English Wikipedia article on the country, the number of Wikipedia language editions with an article on it (Wikidata sitelink count), and the number of Wikidata statements on the country entity. The two families are not interchangeable: because the geographic gap is measured *in English* (language held constant across countries), a residual North/South gap cannot be a language-corpus effect, so H4 isolates the country-representation channel for H1 and the language-corpus channel for H2. All covariate values, official sources, and access dates are versioned in the analysis release.

3.5 Model selection

Fourteen LLMs were benchmarked across five access tiers, spanning roughly two orders of magnitude in parameter count and four training-data geographies (United States, China, Europe, Brazil). **Tier A (open-weight frontier, 5 models)** comprised large open-weight systems accessed via free inference endpoints. **Tier B (open-weight mid, 3 models)** and **Tier C (open-weight small/regional, 2 models)** were run locally via Ollama under pinned model tags; Tier C includes a Brazilian-Portuguese instruction-tuned open model (Cabra-Mistral 7B v3) used for the regional-model test (H3) against a scale-matched globally trained open model. **Tier D (closed accessible, 2 models)** and **Tier E (closed frontier, 1 model)** were accessed via official APIs. Each model was queried with an identical version string or pinned tag throughout the collection window; tags and, where available, weight hashes are recorded in the protocol release to guard against mid-collection model updates. The complete model roster with vendor, parameter count, access route, and execution backend is provided in Supplementary Table S1.

A note on the unit of analysis. We compare *models as served* through a fixed access stack (official vendor API for closed models; a pinned inference endpoint for open-weight models), not idealized model weights in isolation. Provider infrastructure, serving configuration, and API layer are therefore part of what each label denotes, and a residual gap could in principle reflect the stack rather than the weights. We hold this constant by pinning version strings/tags and the maximum-determinism configuration available per provider throughout a fixed collection window, and treat "model" throughout as shorthand for this served-model stack; disentangling weights from infrastructure is a separate question we do not claim to resolve.

3.6 Domain and task taxonomy

The single domain is **air pollution policy**, chosen because it anchors to officially published, verifiable ground truth (national ambient air quality standards, environmental-agency monitoring data, federal control programs) and maps to concrete decision activities of a public environmental manager. Five task types operationalize the domain:

- **T1 — Technical standard.** Recall of binding ambient air quality standards, emission limits, or regulatory thresholds (for example, the annual mean $PM_{2.5}$ standard in force in the country).
- **T2 — Local factual datum.** Recall of officially measured air quality data for a specified city or region and year, as reported by the national or subnational environmental agency.
- **T3 — Health-evidence synthesis.** Synthesis of peer-reviewed epidemiological evidence on the air pollution health burden for the country, scored against a rubric rather than a single value.
- **T4 — Policy instruments.** Identification of policy programs, executing agencies, and legal instruments for air pollution control in force in the country.

- **T5 — Applied recommendation.** Rubric-scored applied recommendation for a defined operational scenario (for example, short-term response to a high-concentration episode under thermal inversion).

T1, T2, and T4 admit verifiable ground truth and carry the primary accuracy signal; T3 and T5 are rubric-scored open-generation tasks. The country-by-task mapping of official source classes, equivalence criteria, and verifiability status is specified in the ground-truth registry (Section 3.9).

3.7 Persona manipulation

Persona is a within-prompt factor with two levels. The **neutral** condition presents the query as a plain information request. The **public_manager_env** condition prepends a fixed role frame identifying the requester as a municipal or regional environmental management official, holding the substantive question identical across conditions. A single role-frame template is applied uniformly across countries and tasks (translated where the prompt language is non-English), so that any between-condition difference is attributable to the persona frame rather than to question wording. Persona order within each prompt-model cell is randomized, and a manipulation check verifies that the role frame is acknowledged in the response. The exact wording, translation procedure, and manipulation-check coding are pre-specified and reproduced in Supplementary Material. Persona is the basis of hypothesis H6, which tests whether the role frame reduces (or amplifies) the country-level accuracy gradient.

3.8 Prompt construction and validation

Prompts were authored for the air pollution policy domain by the research team with domain supervision (PPGSAU/UTFPR) and validated against the ground-truth registry before entering the confirmatory stimulus set. The analysis plan called for a 50-prompt pilot scored by two independent human raters, with Krippendorff's $\alpha \geq 0.70$ required before scaling the design. *That pilot was not run*, and no rater-level α exists for this study. Rubric reliability is instead evidenced after the fact, on the confirmatory data, by a three-vendor judge panel over every item that required a judge (Section 4.11), which returns $\alpha = 0.527$, below the threshold the plan had set. We report this as a departure from the plan rather than as a threshold met. All prompts were authored in English and, for a stratified subset of countries, additionally in a relevant native language (sparse multilingual matrix; Supplementary Section S3); native-language renderings were produced by a high-quality LLM translator (Claude Sonnet 4.6) prompted to preserve the meaning, the persona frame, and proper nouns and acronyms, with the factual ground truth held in English (the target value is language-invariant). We did not employ human native-speaker translation or a back-translation verification step; this is acknowledged as a limitation (Section 5). Following the Brazil pilot validation, every factual claim in a ground-truth entry was required to resolve to an official source URL or a peer-reviewed DOI before the corresponding prompt was admitted, enforcing the project Citation Verification Protocol at the ground-truth generation stage rather than only at the manuscript stage.

3.9 Ground truth and the ground-truth registry

Ground truth for the verifiable tasks (T1, T2, T4) was drawn from official sources of record: national air quality standards and the agencies that issue them (T1), environmental-agency monitoring reports (T2), and federal policy programs with their executing agencies and legal instruments (T4). For T3, the reference is a curated set of peer-reviewed epidemiological studies with resolvable DOIs rather than a single number. Because the validity of the geographic gradient (H1) and the persona effect (H6) depends on ground truth being equally available and equally verifiable across all countries, we constructed a per-country ground-truth registry that, for each (country, task) pair, records the class of equivalent official source, the resolvable URL, the gold value or rubric key, the reference date, the human validator, and a validation flag. The registry design, cross-country equivalence criteria, and the explicit separation of an AI knowledge gap from a ground-truth-availability gap are documented in `docs/ground_truth_registry_design.md` and summarized in Supplementary Material. Entries are versioned by access date, source URL, and a SHA-256 hash of the retrieved document. Where ground truth could plausibly have appeared in training data, a sensitivity subset is constructed from documents dated after each model's training cutoff. Entries whose official source is unavailable or non-equivalent for a country are flagged and excluded from the primary accuracy comparison for the affected (country, task) cell, so that missing ground truth is never scored as a model error.

3.10 Data collection

API and local calls were executed at temperature 0.3 with deterministic seeds where available, with 2 replications per prompt-model-persona combination to estimate stochastic variability within the budget. Collection occurred within a fixed window to minimize model-update confounds, with the pinned tag, version string, and acquisition timestamp recorded for every call; because served-model behaviour can drift as vendors update infrastructure, the reported accuracies are point-in-time estimates relative to this window, and the full per-provider acquisition dates are released with the dataset. All responses were stored alongside complete request metadata (model version string or tag, timestamp, token counts, finish reason, persona condition) in a versioned storage layer with SHA-256 checksums. Response quality gates excluded truncated responses (`finish_reason` $\neq$ `stop`) and responses in a language divergent from the request; both were retained and analyzed separately as secondary outcomes.

Per-cell coverage is not uniform across the model roster: free and rate-limited endpoints (for example GPT-OSS 120B via a shared inference provider) returned fewer admissible responses than the metered APIs, so the scored N ranges from roughly 300 to 780 responses per model. We report each model's realized N alongside its score (Table 1) rather than silently dropping under-covered cells, and a full model $\times$ country $\times$ task coverage matrix, with the quota and exclusion reasons for every shortfall, is provided in Supplementary Material so that no absence is hidden.

3.11 Measures

Scoring assigns each task the most reliable instrument it admits, which is not the same instrument everywhere.

Where the answer is a number in an official register, code decides. For T2 (a measured city-year concentration) and T3 (an attributable-mortality estimate), the returned value is extracted and compared with the authorised range from the register by a deterministic scorer, with plausibility guards that reject implausible extractions — a four-digit year read as a death count, or a national population read as a mortality figure. Comparing a value with a range is exact, and a language model asked to do the same arithmetic can only reproduce it with noise. Code resolves 50.2% of T2 cells and 67.3% of T3 cells; the rest are those where no value could be extracted with confidence.

Where judgement is required, a three-vendor panel decides by its mean. The residual of T2 and T3, together with all of T4, is scored by Gemini 2.5 Pro (Google), Claude Sonnet 4.6 (Anthropic), and DeepSeek-V3 (DeepSeek), chosen for vendor diversity so that intra-vendor correlation does not inflate apparent agreement. The reported score is the mean of the three. T1 and T5 retain the scores of the original single judge (GPT-5-mini), T1 because it always had official ground truth and T5 because it is a rubric judgement against no register.

The original protocol used a single primary judge (GPT-5-mini) for all tasks, which two measurements make untenable. Judges differ substantially in severity, and single-judge reliability is only ICC(2,1)$= 0.56$ against ICC(2,3)$= 0.79$ for the panel mean (Section 4.11); the single judge was also itself among the evaluated models. Reliability is reported three ways: Krippendorff's α (an inter-rater agreement coefficient, $1 =$ perfect, $0 =$ chance), the single-judge ICC(2,1) (intraclass correlation for one rater, two-way random effects), and, as the operative quantity, the **ICC(2,3) reliability of the panel mean**, which by the Spearman-Brown relation exceeds the single-judge value even when individual judges agree only moderately. All three are computed on every item the panel scored rather than on a calibration subset. One panel member (DeepSeek-V3) is also an evaluated model; we test for self-favouring directly and report the result. This study did not include a human gold-standard layer; ground truth was instead anchored to official primary sources through a documentary audit (Section 3.9), and the absence of independent human scoring is stated as a limitation.

The primary composite outcome combines the five pre-specified subcomponents (see Supplementary Section S6), each normalized to $[0, 1]$: (i) factual accuracy against the registry entry (binary T1; partial credit T2–T4); (ii) contextual completeness scored against a pre-validated rubric (T3, T4, 0–5); (iii) citation quality, scored against the DOI- and official-source-verified registry; (iv) hallucination absence, coded binarily (0 hallucinated, 1 faithful); and (v) applied validity, the rubric score for the operational recommendation in T5. The primary composite uses a pre-specified weighting (factual 0.30, contextual 0.25, citation 0.15, calibration 0.15, hallucination 0.15). As a sensitivity analysis we recompute every headline effect under two alternative weightings — equal weights and a PCA-derived weighting (the first principal component of the five subcomponents) — and treat a conclusion as robust only if its direction is stable across all three; all headline effects are (Section 4.12). The composite is computed per (country, model, persona) cell to support the H6 difference-in-differences test.

3.12 Analytical strategy

The three primary confirmatory tests are pre-specified as a single family (F1) and corrected together with the Bonferroni-Holm procedure (a step-down family-wise correction for multiple comparisons). Secondary confirmatory tests (F2) are reported unadjusted within family, and declared exploratory tests (F3) use Benjamini-Hochberg FDR at $q = 0.10$ [37].

H1 (geographic gradient, primary). Country-level mean composite accuracy (n=15) is regressed on the Joshi/HDI gradient via Spearman ρ (rank correlation), pre-specified one-sided with threshold $\rho \geq 0.55$ at $\alpha = 0.05$, with a Mann-Kendall test (a non-parametric monotonic-trend test) as a mandatory non-monotonicity complement and a Global South dummy as a sensitivity contrast.

H4 (corpus-representation mechanism, primary). Country-level partial Spearman ρ relates corpus representation to accuracy after adjusting for HDI and log GDP per capita, using Wikipedia counts as the pre-specified primary proxy ($\rho \geq 0.55$), complemented by exploratory country-coverage proxies (Wikidata sitelinks and statements). Robustness of the supported geographic effects to unmeasured confounding is quantified with E-values [38] (Section 4.12).

H6 (persona effect, primary). The persona effect is tested as a country-level difference-in-differences on the composite outcome: $\Delta_{\mathrm{GS}} = \overline{\mathrm{persona}}_{\mathrm{GS}} - \overline{\mathrm{neutral}}_{\mathrm{GS}}$ and $\Delta_{\mathrm{GN}} = \overline{\mathrm{persona}}_{\mathrm{GN}} - \overline{\mathrm{neutral}}_{\mathrm{GN}}$, with H6 statistic $\Delta_{\mathrm{GS}} - \Delta_{\mathrm{GN}}$. H6 is supported (one-sided, $\alpha = 0.05$) if the persona condition reduces the Global South gap by at least 0.05 on the $[0, 1]$ composite (equivalently, 5 percentage points of absolute accuracy). Inference uses a country-level permutation test (5,000 permutations) reported alongside a parametric paired test; the two-sided test is reported as a secondary check because persona amplification is theoretically possible and is reported transparently if observed.

H3 (regional model, secondary). H3 is tested with a pre-specified one-sided Mann-Whitney contrast (Cliff's δ effect size) comparing Cabra-Mistral 7B v3 against the other models on the air pollution policy tasks.

H2 and H5 (exploratory). The prompt-language $\times$ country interaction (H2) and the open-frontier versus closed-frontier contrast (H5) are reported descriptively, with formal tests under F3 correction where applied.

As a cross-cutting corroboration of the tier gap, a Gaussian linear mixed model (the `statsmodels` MixedLM estimator, REML) regresses the composite on the Global South indicator (adjusting for centered HDI) with crossed random intercepts for country and model, with country-only and model-only single-grouping fits as numerical checks. Pre-specified robustness analyses include leave-one-country-out re-estimation, a range-restriction check, and the composite-weighting sensitivity described above (Section 4.12); robustness of the supported geographic effects to unmeasured confounding is summarized with E-values. We also re-estimate the Global South effect in a Bayesian hierarchical model (`pymc` [39], weakly informative priors, crossed country and model intercepts), and probe the corpus mechanism with an exploratory country-level mediation model (`semopy` [40]; HDI $\rightarrow$ country-coverage (sitelinks) $\rightarrow$ accuracy, with a bootstrap confidence interval on the indirect effect). The persona manipulation is characterized by a role-frame-acknowledgement check on the response text. An *a priori* power analysis at the planning stage, under pilot-informed effect sizes, indicated adequate family-wise power for the primary trio under Bonferroni-Holm and a minimum detectable persona gap reduction of a few percentage points; power details are in the pre-specified analysis plan.

3.13 Study protocol and open science

The full protocol — hypotheses, prompts with hashes, the ground-truth registry, the persona template, scoring rubrics, mixed-model and DiD specifications, multiple-comparison corrections, and exclusion criteria — is pre-specified in advance of confirmatory data collection. The analytic dataset, after quality gates but before model fitting, is frozen and hashed so that the confirmatory analysis can be audited against the locked specification. The full protocol, prompts, ground-truth references, analysis code, and anonymized response data will be released on publication under CC-BY-4.0 (data and prompts) and the MIT License (code) [41]; a companion Data Paper is prepared for *Scientific Data*.

3.14 Ethical considerations

No personal data from end-users of LLMs were collected. The study analyzes model outputs to publicly relevant policy questions; domain experts who contribute to ground-truth validation are acknowledged as co-authors when they meet ICMJE criteria. LLM provider terms of service were reviewed and respected, and responses are redistributed solely for research under fair-use and reproducibility provisions, excluded from any commercial application.

4 Results

Sample note. We report a pre-specified *15-country* benchmark and its *post-hoc extension to 25 countries* (the latter adds six Global North references — the United Kingdom, Canada, Australia, South Korea, France, Italy — and four native-pair countries — Colombia, Chile, Portugal, Angola). Fourteen LLMs are scored across five tasks, two personas, and four languages (English plus a country-native language — Spanish, Portuguese, or Hindi — for the nine applicable countries, giving the within-country English/native pairs that test H2). Every effect computed on the 25-country sample is reported alongside its pre-specified 15-country value; no pre-specified conclusion is revised on the extension alone. Scoring is adjudicated by code wherever the answer is a number checkable against an official register, and by the mean of a three-vendor judge panel elsewhere (Section 4.11); the T1 ground truth is anchored to official primary sources for all 25 countries (Section 4.9).

4.1 Sample and scoring

The analysed corpus comprises 9,251 scored responses on the $[0, 1]$ composite, of which 7,580 are English-prompt responses across the 25 countries (the remainder are the within-country native-language pairs used for H2).

Scoring is assigned by the most reliable instrument each task admits. For T2 and T3, where the correct answer is a number published in an official register, the verdict is computed by code against that register: comparing a returned value with an authorised range is exact, and a language model asked to do the same arithmetic can only reproduce it with noise. Code resolves 50.2% of T2 and 67.3% of T3; the residual, together with all of T4, is scored by the mean of a three-vendor panel (Gemini 2.5 Pro, Claude Sonnet 4.6, DeepSeek-V3). T1 and T5 retain their original scores, T1 because it always had official ground truth and T5 because it is a rubric judgement with no register to check against. In total 4,475 cells carry a corrected score.

The panel mean replaces the single-judge design of the original protocol, in which the judge (GPT-5-mini) was itself among the evaluated models. Two measurements justify the change. Judges differ sharply in severity — mean composite 0.617 for Gemini, 0.451 for Claude, 0.379 for DeepSeek — so a single judge fixes an arbitrary level; and single-judge reliability is only $\mathrm{ICC}(2,1) = 0.56$ against $\mathrm{ICC}(2,3) = 0.79$ for the panel mean (Section 4.11). One panel member (DeepSeek-V3) is also an evaluated model, so we checked directly for self-favouring: DeepSeek scores its own outputs *more* harshly relative to the other two judges (-0.177) than it scores any other model's (-0.123 to -0.160), so its own ranking is not an artefact of judging itself.

4.2 Accuracy by model

Table 1 orders the 14 models by mean composite. Frontier systems lead and small open-weight models trail. The result bearing on H3 is unambiguous: the Brazilian-Portuguese regional model Cabra-Mistral 7B is the *weakest* of all models (0.351 versus 0.546 for the rest; Mann-Whitney $p \approx 0$, Cliff's $\delta = -0.45$, a non-parametric effect size in $[-1, 1]$), contradicting the optimistic H3 framing that a regional model buys domain accuracy. Scale and frontier training dominate regional instruction tuning at the 7B level.

Table 1: Composite accuracy by model across 25 countries (code-adjudicated where the answer is a register value, panel mean elsewhere; $[0, 1]$). N = scored English-prompt responses.

Model	N	Mean
DeepSeek-V3	500	0.675
GPT-5	780	0.659
Gemini 2.5 Flash	490	0.620
GPT-5-mini	777	0.615
Claude Haiku 4.5	490	0.577
Qwen3 32B	498	0.551
Command R+	500	0.520
Llama 3.3 70B	474	0.508
Llama 4 Scout	500	0.483
GPT-OSS 120B	298	0.478
Phi-4 14B	500	0.473
Qwen3 14B	500	0.473
Llama 3.1 8B	777	0.438
Cabra-Mistral 7B	496	0.351

4.3 Accuracy by task: the recall floor

Difficulty is sharply task-dependent (Table 2). The hardest task by a wide margin is **T1 (technical-standard recall, 0.367)** — the task that requires returning the national annual $PM_{2.5}$ standard in force. The open recommendation task (T5) scores highest (0.773) because its rubric rewards plausible, well-structured policy advice that does not hinge on a single fact. Pooling the two recall tasks (T1, T2; mean 0.413) against synthesis and recommendation (T3–T5, 0.619), the difference is large (Mann-Whitney $p \approx 0$, Cliff's $\delta = -0.47$). LLMs are weakest exactly where applied environmental management needs them most: recalling the precise regulatory threshold in force.

The correction of the ground truth is itself informative here, and provides an internal check on it. T1 always had official ground truth, and its score does not move at all (0.367 before and after). T2 was previously scored against a placeholder, and it rises from 0.368 to 0.460 once each returned concentration is compared with the authorised range by code. The correction therefore moved exactly the task whose ground truth was defective and left untouched the task whose ground truth was sound, which is the behaviour expected if the correction is right. It also revises the earlier reading: the floor is real but shallower than reported, and it is concentrated in normative recall (T1) rather than shared equally with local-datum recall (T2).

Table 2: Composite accuracy by task type (25 countries). T2 and T3 are code-adjudicated where the returned value is checkable against the official register; T4 is panel-scored; T1 and T5 retain their original scores.

Task	N	Mean
T5 (applied recommendation)	1,508	0.773
T3 (health-evidence synth.)	1,519	0.549
T4 (policy instruments)	1,494	0.534
T2 (local factual datum)	1,530	0.460
T1 (technical standard)	1,529	0.367

4.4 H2 — asking in the local language lowers accuracy

This is the study's strongest and most directly actionable result.

For the nine countries with a target native language (Brazil/Portugal/Angola → Portuguese, Mexico/Argentina/Peru/Colombia/Chile → Spanish, India → Hindi), each prompt is rendered in both English and the native language, holding content fixed, so the English/native contrast is within-country and within-model. Replicates are averaged within each (model, prompt) cell before pairing, because pairing replicate-to-replicate would inflate n by the number of replicates and make the test anticonservative. Across the 839 matched cells, native-language prompting *lowers* mean composite accuracy by **5.0 percentage points** (Wilcoxon signed-rank $p < 10^{-5}$).

The penalty tracks the resource level of the language, and all three are now significant (Table 3): Hindi −12.0 pp ($p = 8 \times 10^{-5}$), Spanish −4.6 pp ($p < 10^{-5}$), Portuguese −3.9 pp ($p = 8 \times 10^{-5}$). India, which scores highest of all 25 countries in English, drops the furthest in Hindi. The effect survives every alternative weighting of the composite (−4.9 pp under equal weights, −5.8 pp on the factual subcomponent alone), so it is not an artefact of how the five subcomponents are combined.

The applied implication is direct, and it is the opposite of the intuition. A manager who suspects an English-language model serves their country poorly, and responds by asking in the national language, makes the answer *worse* — and worse by the largest margin precisely where the language is least represented in training corpora. Taken with H6 below and the regional-model result above, all three localisation remedies fail together, and this one actively harms. This is consistent with the linguistic-resource hierarchy [10] and with the language-corpus channel of H4.

Table 3: H2: native-minus-English accuracy by native language (paired within country and model, replicates averaged within cell, 25-country sample).

Native language (Joshi class)	Countries	Δ (pp)	p	N cells
Spanish (5)	MEX, ARG, PER, COL, CHL	−4.6	$< 10^{-5}$	457
Portuguese (4)	BRA, PRT, AGO	−3.9	8×10^{-5}	311
Hindi (4)	IND	−12.0	8×10^{-5}	71
Overall		−5.0	$< 10^{-5}$	839

4.5 H6 — persona does not narrow the geographic gap

The persona factor was fully crossed, so H6 is tested as a difference-in-differences. The public-environmental-manager frame leaves the North/South gap essentially unchanged: +4.3 pp under the neutral frame versus +4.8 pp under the manager frame, a DiD of +**0.5** pp, far below the pre-specified 5 pp threshold; a country-level permutation test (5,000 permutations of the tier label) gives $p = 0.22$. **H6 is not supported**: presenting the query as coming from a public environmental manager neither narrows nor amplifies the Global South disadvantage, consistent with evidence that expert personas do not reliably improve factual accuracy [31, 32]. A manipulation check finds that 50.2% of persona-condition responses explicitly acknowledge the role frame, so the null is not an artefact of the frame being universally ignored.

4.6 H1 — a tier gap, and an unresolved gradient

H1 was pre-specified in two parts, and they now separate cleanly: one holds and one does not.

The tier gap holds. The ten Global North countries average 0.563 against 0.518 for the fifteen Global South countries, a gap of +4.6 **percentage points** whose 95% bootstrap confidence interval over countries is $[+1.2, +7.8]$ pp and *excludes zero*. It is stable to dropping any single country (leave-one-out range $[+4.0, +5.5]$ pp) and to the composite weighting (+3.8 pp under equal weights, +7.7 pp on the factual subcomponent alone, where it widens).

The monotonic gradient does not. At $n = 25$, Spearman $\rho(\text{accuracy}, \text{HDI}) = +0.36$ does not reach significance ($p = 0.080$), and neither does the Mann-Kendall trend ($p = 0.072$). At the pre-specified $n = 15$ it is essentially absent ($\rho = +0.08$). No leave-one-country-out re-estimate reaches the pre-specified $\rho \geq 0.55$ threshold (range $[+0.29, +0.48]$), and no alternative weighting does either ($\rho = +0.31$ under both equal weights and the factual subcomponent). We therefore report a **tier difference we can demonstrate and a development gradient we cannot**: the disadvantage is established as a difference between groups, while its shape across the development range remains exploratory. Under Bonferroni-Holm across the primary family {H1-gradient, H4, H6}, no test rejects its null.

The country ordering also retains large axis-orthogonal heterogeneity (Table 4): India (0.650, Global South) tops all countries, South Africa (0.580) and Indonesia (0.578) outrank most of the Global North, and Canada (0.506) and Germany (0.537) sit low. The disadvantage is a tier difference layered over substantial country-specific variation, not a clean ranking. This heterogeneity is itself part of why the gradient does not resolve.

Table 4: Composite accuracy by country, 25-country sample (tier: GN = Global North, GS = Global South). Ordered by mean.

Country	Tier	Mean	Country	Tier	Mean
India	GS	0.650	Bangladesh	GS	0.536
Japan	GN	0.603	France	GN	0.532
South Korea	GN	0.592	Peru	GS	0.526
Australia	GN	0.590	Colombia	GS	0.524
Italy	GN	0.581	Chile	GS	0.510
South Africa	GS	0.580	Canada	GN	0.506
United States	GN	0.579	Mexico	GS	0.502
Indonesia	GS	0.578	Kenya	GS	0.500
United Kingdom	GN	0.566	Angola	GS	0.482
Philippines	GS	0.554	Brazil	GS	0.469
Portugal	GN	0.549	Nigeria	GS	0.458
Germany	GN	0.537	Egypt	GS	0.456
			Argentina	GS	0.442

4.7 H4 — corpus representation: mechanism unresolved

The pre-specified mechanism for the geographic gap was training-corpus representation, with the pre-specified proxy being Wikipedia article/edition count. That proxy is **null**: Spearman $\rho = +0.09$ ($p = 0.65$) uncontrolled, and it remains null partialling out HDI and log GDP per capita, which is the adjustment set the plan specified. The two control sets give the same answer because they carry nearly the same information: HDI and log GDP per capita correlate at $\rho = 0.97$ across these 25 countries. **H4 as pre-specified is not supported.**

We then explored, post hoc, whether a different channel is at work. Because the gap is measured *in English*, a residual North/South difference cannot reflect how much text exists in a country's language, so we distinguish language-corpus size from how broadly the corpus is *about* a country. The best of three country-coverage proxies — the number of Wikipedia language editions with an article on the country (Wikidata sitelinks) — correlates with accuracy at $\rho = +0.35$, which does not reach significance ($p = 0.087$), and attenuates further after adjusting for HDI (partial $\rho = +0.17$, $p = 0.44$).

We therefore report the mechanism as **unresolved**. Neither channel is established at country level: language-corpus size is null, and country coverage is suggestive at best. Two features of the design limit what can be concluded rather than what is true. The Joshi variable takes only three values across our 25 countries and eighteen of them share Class 5, so a rank correlation on it has almost no variance to work with; and the pre-specified corpus proxy assigns the size of the dominant official language's Wikipedia edition, taking only eleven distinct values for 25 countries, nine of them tied on the English edition. The honest reading is that this design *cannot separate* the channels at country level, not that it ruled either out.

The one place the corpus channel does show itself is H2, where the unit is the language rather than the country. There the direction is as predicted, though with only three native languages it remains descriptive: the per-language penalty falls monotonically with mC4 token volume [15] and OSCAR size [16] (Spearman = -1.00 over the three), with Hindi, the smallest corpus (mC4 24B tokens), carrying the largest penalty. That the corpus account survives where the measurement has resolution, and fails where it does not, is consistent with an instrument problem rather than an absent mechanism — but consistency is not evidence, and we do not claim the mechanism.

4.8 H5 — open versus closed-accessible

Grouping by access type, closed-accessible models average +13.0 pp above open-weight models. This runs against the optimistic H5 framing; the open arm includes larger models (Llama 3.3 70B, DeepSeek-V3), so the gap is not attributable to scale alone, although DeepSeek-V3 itself now ranks first overall. H5 is interpreted descriptively.

4.9 Ground-truth provenance

Because measured "accuracy" depends on the ground truth, the technical-standard target (T1) for all 25 countries was anchored to official primary sources by documentary verification rather than expert assertion, and is reproducible from the cited URLs. Most countries have a value read from the official regulation (e.g. Brazil's annual $PM_{2.5}$ standard of 17 μg/m^3 read verbatim from Annex I of CONAMA Resolution 506/2024 in the Diário Oficial da União; Colombia's 25 μg/m^3 read from Tabla 1 of Resolución 2254/2017). Three countries have *no* national $PM_{2.5}$ standard, confirmed from the official text — Nigeria's NESREA regulation sets PM10 only, Argentina has no binding federal value, and Angola has no national air-quality legislation — a fact the audit established against circulating but incorrect secondary figures, and in which a faithful model must decline to state a non-existent threshold rather than fabricate one.

The same standard was extended to the tasks that previously lacked it. T2 is adjudicated against WHO Ambient Air Quality Database v6.1 (23 of 25 countries have a city in the base; Angola has none and Cairo carries no $PM_{2.5}$ value), T3 against WHO GHO indicator AIR_41 with its uncertainty intervals, and T4 against Appendix 1 of the UNEP Global Assessment of Air Pollution Legislation, covering 24 of 25 countries. This provenance is more reproducible than expert validation: any reader can re-verify each value from the official source.

4.10 Qualitative failure modes

To make the composite numbers concrete, Box 1 shows three representative failures drawn verbatim from the response logs (run IDs available in the release).

The first mode is the most consequential for the no-standard cases (Angola, Nigeria, Argentina): rather than abstaining, models invent a plausible regulatory cutoff and disagree across replicates and models on the invented value — exactly the behaviour a public manager cannot detect without the official source.

4.11 Inter-judge reliability

Reliability is reported on the sample that produces the effects, not on a calibration subset. All 3,190 items that required a judge were scored by all three panel members (Gemini 2.5 Pro, Claude Sonnet 4.6, DeepSeek-V3), giving Krippendorff's $\alpha = 0.527$ (interval), single-judge $\mathrm{ICC}(2,1) = 0.558$, and — the operative quantity, since the analysis uses the panel mean — $\mathrm{ICC}(2,3) = \mathbf{0.791}$.

Table 5: Box 1 — Representative failure modes (verbatim, lightly truncated).

Failure mode	Example
Fabricating a non-existent standard (country with no national $PM_{2.5}$ norm)	For **Angola** (no national standard), Gemini 2.5 Flash asserted a "25 μg/m^3" annual standard in one replicate and "15 μg/m^3" in another; GPT-5 asserted "15 μg/m^3" for **Nigeria**, which regulates only PM_{10} (NESREA). A faithful response would decline to state a value that does not exist.
Missing the binding value (T1 floor)	For **Brazil** (official annual standard 17 μg/m^3, CONAMA 506/2024), the modal model answer was 15 μg/m^3 (42 responses), with 10, 20, and 25 also common; the official value was almost never returned.
Native-language degradation (H2)	For **India** (annual standard 40 μg/m^3), GPT-OSS 120B returned the correct value in English but an *empty* answer to the same question in Hindi.

Pairwise Pearson agreement is +0.80 between Claude and DeepSeek and +0.65–0.66 between each and the systematically more lenient Gemini.

Two points deserve emphasis because they cut against the original design. First, a single judge would carry $\mathrm{ICC}(2,1) = 0.56$, which is moderate at best; the published protocol relied on one, and the panel mean is what makes the composite defensible. Second, these values are *lower* than the $\mathrm{ICC}(2,4) = 0.89$ obtained on the 131-item calibration sample reported in earlier drafts. The calibration sample was stratified and small; measured on 24 times as many items, and on the same items that produce the effects, agreement is more modest. We report the larger, less flattering estimate because it is the one that describes the instrument actually used.

During the re-scoring we also identified, and corrected, a defect of our own making. An early version of the T2 prompt asked the judge to compare a returned concentration against a tolerance band expressed in words, which required the judge to perform the arithmetic. Inter-judge concordance on those items fell to $r = -0.04$. Pre-computing the admissible range and asking only whether the value falls inside it raised concordance on the same items to $r = +1.00$. The disagreement was an artefact of our prompt, not a genuine divergence of judgement, and the 274 scores produced under the ambiguous wording were discarded and recollected. Arithmetic belongs to code; the judge should be asked only what requires judgement.

4.12 Robustness

Influential observations. Leave-one-country-out re-estimation leaves the tier gap in $[+4.0, +5.5]$ pp, so no single country drives it. The HDI gradient ranges over $[+0.29, +0.48]$ and never reaches the pre-specified 0.55 threshold under any drop, which is part of why we report it as exploratory.

Metric robustness. The composite blends five subcomponents, so one might worry it dilutes or manufactures the effects. It does not change the conclusions. Under equal weights the tier gap is +3.8 pp, the native-language penalty −4.9 pp, the recall floor $\delta = -0.43$ and the gradient $\rho = +0.31$; restricting to the single most objective subcomponent — factual accuracy against the register value, ignoring completeness, calibration, and style — gives +7.7 pp, −5.8 pp, $\delta = -0.42$ and $\rho = +0.31$, with the recall floor deepening (T1+T2 factual accuracy 0.345 versus 0.648 for T3–T5). The two supported effects hold in every weighting and the gradient fails in every weighting, so neither conclusion is an artefact of how subcomponents are mixed.

Range extension. The ten countries added post hoc fall *within* the HDI range already spanned by the original fifteen ($[0.548, 0.950]$, from Nigeria to Germany), so the extension increases the density of observations rather than the range of the predictor.

Model-based corroboration, and where it disagrees. Two model-based re-estimations on the corrected scoring qualify the tier gap rather than simply confirming it, and we report the disagreement.

A Gaussian linear mixed model with crossed random intercepts for country and model, adjusting for centred HDI, estimates a Global South penalty of $\beta = -0.060$ (SE 0.037) that does *not* reach significance ($p = 0.099$). A Bayesian hierarchical re-estimation with weakly informative priors agrees in magnitude and is more decisive about the sign: posterior mean -0.047, 94% HDI $[-0.085, -0.006]$, $P(\beta < 0) = 0.984$, with clean convergence ($\hat{R} = 1.000$ over four chains).

So two of the three ways of asking the question — the country-level bootstrap and the Bayesian model — put the interval clear of zero, while the frequentist mixed model leaves it marginal. The estimates agree on magnitude (roughly -5 to -6 percentage points) and on sign; they disagree on whether that is distinguishable from no effect at conventional thresholds, because the mixed model charges the substantial between-country heterogeneity (Table 4) to the error term. We therefore describe the tier gap as supported but *modest and specification-sensitive*, which is a weaker claim than earlier drafts made on the single-judge scores ($\beta = -0.077$, $p = 0.007$).

Where the tier gap lives. The composite averages five tasks, and the gap is not spread evenly across them. Re-estimating the same contrast as a binomial mixed model on each task separately — outcome coded as returning the register value, with crossed random intercepts for country and model — shows the disadvantage scaling with how much the task depends on a fact specific to that country (Table 6). On the binding national standard (T1) the Global South odds of a correct answer are 0.23 of the Global North's (52.0% versus 29.4% raw), and on policy instruments (T4) 0.34; on health-evidence synthesis (T3), where the answer is an international estimate rather than a national one, the ratio rises to 0.79; and on open recommendation (T5), the one task with no register value to get right, the gap disappears entirely.

This is the most important qualification of the tier gap in either direction. The gap is not weak — it is *concentrated*, and the composite dilutes it by averaging the tasks where it is severe with a task where it does not exist. On the question a public manager most often brings to these systems, what is the standard in force here, a Global South country is served at roughly a quarter of the odds of a Global North one.

Table 6: Global South versus Global North odds of returning the register value, by task. Binomial mixed model, crossed random intercepts for country and model.

Task	Answer is	GN	GS	OR	95% interval
T1 (technical standard)	a national value	52.0%	29.4%	0.23	$[0.19, 0.28]$
T4 (policy instruments)	national instruments	63.1%	48.5%	0.34	$[0.28, 0.41]$
T2 (local measured datum)	a national value	45.2%	35.6%	0.59	$[0.51, 0.68]$
T3 (health-evidence synth.)	an international estimate	60.6%	56.2%	0.79	$[0.69, 0.91]$
T5 (applied recommendation)	no register value	99.6%	99.7%	2.67	$[0.80, 9.00]$

E-values. On the risk-ratio scale, the tier gap carries an E-value of 1.65 at the point estimate and 1.25 at the confidence limit nearest the null. The second number is the operative one [38]: an unmeasured country-level confounder associated with both tier and accuracy at RR ≈ 1.25 would suffice to render the gap compatible with no effect. That is a low bar, and it belongs in the reader's assessment.

We do *not* report an E-value for the HDI gradient, although earlier drafts did (5.4). The gradient does not reach significance ($p = 0.080$), so its interval already includes the null and the E-value at that limit is 1.00 — no confounder at all is required. Reporting only the point-estimate E-value would imply a robustness the estimate does not have. The conversion is also a poor fit for a continuous, country-level exposure, which the risk-ratio formulation assumes away.

4.13 Summary

Table 7 maps each hypothesis to its finding, the candidate mechanism, and an explicit evidence status, so that no claim is read as stronger than the data support.

Table 7: Hypotheses, findings, candidate mechanisms, and evidence status.

Hypothesis	Finding	Candidate mechanism	Evidence status
H2 (language)	-5.0 pp; Hindi -12.0	language-corpus size	Supported; mechanism descriptive ($n = 3$)
H3 (regional model)	$\delta = -0.45$	scale > regional tuning	Supported
H1 (tier gap)	+4.6 pp, CI excludes zero	tier (development)	Supported; modest and specification-sensitive
H1 (gradient)	$\rho = 0.36$, $p = 0.080$	development (HDI)	Exploratory; not significant
H4 (geographic mech.)	sitelinks $\rho = 0.35$, $p = 0.087$	country coverage	Exploratory; unresolved
H4 (pre-specified)	Wikipedia-size null	language-corpus size	Not supported
H5 (open/closed)	+13.0 pp closed	access type / scale	Descriptive
H6 (persona)	DiD +0.5 pp	—	Not supported

Correcting the ground truth reorders what this study can claim. **Supported, with effect sizes:** a native-language *penalty* of -5.0 pp (Wilcoxon $p < 10^{-5}$, $n = 839$ cells), significant in all three languages and largest for Hindi (-12.0 pp); a *recall floor* on the tasks that require returning a binding value (T1+T2 versus others, $\delta = -0.47$, concentrated in T1 at 0.367); the regional model as the *worst* of fourteen ($\delta = -0.45$); and a Global North/South *tier* gap of +4.6 pp (bootstrap 95% CI $[+1.2, +7.8]$, excludes zero), modest and sensitive to specification on the composite — the Bayesian re-estimation puts it clear of zero ($P(\beta < 0) = 0.984$) while the frequentist mixed model leaves it marginal ($p = 0.099$), and its E-value at the confidence limit is only 1.25 — but *unambiguous where it matters*: on returning the binding national standard, Global South odds are 0.23 of Global North odds ($[0.19, 0.28]$), and the gap scales with how far the answer depends on a country-specific fact, vanishing on the one task with no register value to get right. **Exploratory or not supported:** the development *gradient* does not reach significance ($\rho = 0.36$, $p = 0.080$; Mann-Kendall $p = 0.072$; $\rho = 0.08$ at the pre-specified $n = 15$), so the tier difference is demonstrated but its shape is not; the mechanism is unresolved, with the pre-specified language-corpus proxy null ($\rho = 0.09$) and country coverage no longer significant ($\rho = 0.35$, $p = 0.087$; partial $p = 0.44$); and persona does *not* narrow the gap ($p = 0.22$).

The centre of gravity is therefore practical rather than aetiological. What this study demonstrates is that the three remedies an agency would actually reach for — ask in the national language, declare the official role, procure a regionally tuned model — all fail, and that the first makes matters materially worse; and that models are least reliable precisely on the recall of the binding value in force. Why the tier gap exists, we do not resolve. This matters for environmental management in the Global South, where the disease burden is large [5, 42] and where exactly the hardest task — recalling the binding standard — is the one a manager most needs [10, 12].

5 Discussion

We set out to measure whether LLMs answer air pollution policy questions as reliably for the Global South as for the Global North, whether prompt framing changes the answer, and what mechanism drives any gap. The benchmark — 25 countries, 14 models, four languages, five tasks, two personas, scored against official ground truth by code where the answer is a register value and by a three-vendor panel elsewhere — returns a disciplined and partly counter-intuitive picture. The harms we can demonstrate are practical rather than aetiological: the three remedies an agency would reach for all fail, the most intuitive of them makes the answer materially worse, and models are least reliable on the recall of the binding value in force. A North/South tier difference persists; what that difference is made of, we do not resolve.

5.1 Three intuitive remedies, all failing — and one that harms

The applied contribution is largely negative, and it is the study's most actionable result.

The local language does not help and materially hurts. Asking in the country's own language lowers accuracy by 5.0 pp (Wilcoxon $p < 10^{-5}$, $n = 839$ matched cells), significantly in all three languages, and the penalty scales with the scarcity of the language's corpus: Hindi −12.0 pp, Spanish −4.6, Portuguese −3.9. India, the highest-scoring country of all 25 in English, falls furthest in Hindi. This is the mechanism the resource hierarchy predicts [10, 15], and here the measurement has the resolution to see it, because the unit is the language rather than the country.

The local-manager persona does not narrow the gap (DiD +0.5 pp, permutation $p = 0.22$), consistent with evidence that expert personas do not reliably improve factual accuracy [31, 32]. A manipulation check confirms half of the responses explicitly take up the role frame, so the null is not a failure of the manipulation.

The regional model — Brazilian-Portuguese Cabra-Mistral 7B — is the *worst* performer of all fourteen ($\delta = -0.45$), so reaching for a locally tuned small model trades away the scale and frontier training that actually carry domain accuracy.

Practitioners deploy all three of these as fixes; none survives contact with the data, and the one a manager is most likely to try first is the one that does active damage. The honest message for Global South environmental agencies is that no cheap prompt- or model-level workaround substitutes for verifying the binding value against the official source.

5.2 The recall floor sits on the values that bind

Accuracy collapses on the tasks that require returning a specific value in force: 0.413 for normative and local-datum recall against 0.619 for synthesis and recommendation ($\delta = -0.47$), and the floor is deepest on the national standard itself (T1, 0.367). The risk is concentrated rather than diffuse. A manager drafting prose about air quality policy is comparatively well served; a manager retrieving the regulatory cutoff that a decision turns on is in the worst cell of the design, and worse still if they ask in a lower-resource language.

That this floor is genuine, and not an artefact of how we scored it, is supported by an internal check. Correcting the ground truth raised T2 — the task that had been scored against a placeholder — from 0.368 to 0.460, while leaving T1 — which always had official ground truth — at exactly 0.367. The correction moved the task whose key was defective and left untouched the task whose key was sound.

5.3 A tier difference we can show, and a gradient we cannot

H1 was pre-specified in two parts, and the corrected scoring separates them: the **tier gap** holds and the **monotonic gradient** does not.

The gap is +4.6 pp with a bootstrap CI of $[+1.2, +7.8]$ that excludes zero, stable to dropping any country and to every composite weighting, widening to +7.7 pp on the factual subcomponent alone. It is nonetheless modest and sensitive to how the question is asked. A Bayesian hierarchical model puts the effect clear of zero (-0.047, 94% HDI $[-0.085, -0.006]$, $P(\beta < 0) = 0.984$), while a frequentist mixed model with crossed random intercepts leaves it marginal ($\beta = -0.060$, $p = 0.099$), because it charges the large between-country heterogeneity to the error term. The three approaches agree on magnitude and sign and disagree on significance, and the E-value at the confidence limit nearest the null is only 1.25, so a country-level confounder of modest strength would suffice to explain the gap away.

That modesty, however, is an artefact of the composite. Estimated task by task on the binary outcome of returning the register value, the gap scales with how much the answer depends on a fact specific to that country: odds ratio 0.23 ($[0.19, 0.28]$) on the binding national standard, 0.34 on national policy instruments, 0.59 on the local measured datum, 0.79 on an international health estimate, and no gap at all on open recommendation, the one task with no register value to get right. The disadvantage is not weak but *concentrated*, and averaging five tasks into one composite dilutes it with tasks where it does not exist. On the question a manager actually brings to these systems — what is the standard in force here — a Global South country is served at roughly a quarter of the odds of a Global North one. We regard this as the more faithful statement of the geographic finding than the composite gap, and it is the reading we carry into the implications.

The gradient, by contrast, fails on every reading: $\rho = 0.36$ with HDI at $n = 25$ does not reach significance ($p = 0.080$), the Mann-Kendall trend agrees ($p = 0.072$), no leave-one-out estimate approaches the pre-specified 0.55 threshold, and at the registered $n = 15$ it is essentially absent ($\rho = 0.08$).

We report this asymmetry plainly because it is the honest description: a difference between groups that we can demonstrate, and a shape across the development range that we cannot. The country ordering shows why. India tops all 25 countries despite being Global South; South Africa and Indonesia outrank most of the Global North; Canada and Germany sit low. The disadvantage is a tier difference layered over substantial country-specific idiosyncrasy, and that idiosyncrasy is precisely what a rank correlation over 25 countries lacks the resolution to see through.

5.4 The mechanism is unresolved

Our pre-specified mechanism test fails: the language-corpus proxy is null ($\rho = 0.09$), so H4 as registered is not supported. The post-hoc alternative does not rescue it. Because the gap is measured *in English*, a residual North/South difference cannot be an effect of how much text exists in a country's language, which made country-coverage breadth the natural candidate; but Wikidata sitelinks correlate at only $\rho = 0.35$ ($p = 0.087$) and attenuate further after adjusting for development (partial $\rho = 0.17$, $p = 0.44$).

Neither channel is established at country level, and we say so rather than promoting the surviving correlation. Two features of the design bound what can be concluded rather than what is true: the Joshi variable takes three values across 25 countries with eighteen sharing one class, and the pre-specified corpus proxy takes eleven distinct values with nine tied on the English edition. The instrument lacks the resolution to separate the channels at country level. That the corpus account does show itself in H2, where the unit is the language and the variation is real, is consistent with an instrument problem rather than an absent mechanism — but consistency is not evidence, and we do not claim the mechanism.

5.5 The methodological lesson: the key matters more than the sample

Earlier drafts of this study drew a different conclusion from the same responses, and the reason is worth stating, because it generalises beyond this paper.

The original scoring compared model answers for two of the five tasks against placeholder ground truth, and delegated to a language model the arithmetic of deciding whether a returned number fell inside an authorised range. Rebuilding those keys from official registers (WHO Ambient Air Quality Database v6.1, WHO GHO AIR_41, UNEP Global Assessment of Air Pollution Legislation) and moving the comparison into code changed the substantive conclusions: the development gradient fell from $\rho = 0.51$ ($p = 0.004$) to $\rho = 0.36$ ($p = 0.080$), while the native-language penalty more than doubled, from -2.1 pp to -5.0 pp, and turned Spanish from a null into a significant harm. The dataset did not change. Only the measuring instrument did.

The field's instinct when a geographic-bias effect is fragile is to add countries. Our experience is that the ground truth deserves the scrutiny first. A defective key does not simply add noise; it adds *structured* noise, because the defects concentrate wherever official data are hardest to obtain, which is exactly where the hypothesis lives. We also found that an ambiguous scoring prompt can masquerade as genuine inter-judge disagreement: concordance on the affected items was $r = -0.04$ under our wording and $r = +1.00$ once the admissible range was pre-computed and the judge was asked only whether the value fell inside it. Arithmetic belongs to code; a judge should be asked only what requires judgement.

5.6 Why the domain matters

The stakes are not benchmarking abstractions. Fine particulate matter is among the leading environmental risk factors for premature mortality, and the burden falls disproportionately on the Global South. The World Health Organization attributes 88,548 deaths in Brazil in 2021 to ambient air pollution (uncertainty interval 69,477 to 108,431), led by ischaemic heart disease, stroke and acute lower respiratory infections [43]; short-term exposure continues to drive measurable mortality in Brazilian cities [42]. When a manager delegates a normative lookup or an episode recommendation to an LLM, an inaccurate answer can propagate into a regulatory brief or an emergency response.

It would be natural to close this argument by saying that the model is least reliable exactly where the consequences are greatest. We do not make that claim, because we did not measure it: disease burden is not among our country covariates, and no test in this study relates accuracy to mortality. The one country-level observation we do have points the other way, since India carries one of the largest $PM_{2.5}$ mortality burdens in the world and is also the highest-scoring country in our sample. Whether reliability is misaligned with need is a well-posed question, and a straightforward one to answer by adding an age-standardised attributable-mortality covariate, but it is not a question this design answers. We read the pattern we *did* measure within the coloniality-of-knowledge framework [12]: LLMs are compressed summaries of corpora produced under uneven digital and publication access, and the language penalty is a measurable link from corpus asymmetry to knowledge asymmetry — the one link of the three we set out to test where the measurement has the resolution to see it.

5.7 Boundary conditions and opportunities

The findings rest on a defensible base for an LLM benchmark: ground truth anchored to official primary sources for all 25 countries, code adjudication wherever the answer is a checkable register value, and a three-vendor panel elsewhere whose reliability is reported on the same 3,190 items that produce the effects ($\mathrm{ICC}(2, 3) = 0.79$). The items below map to concrete, pre-scoped next studies that the released protocol already supports.

From panel-scored to gold-anchored. Single-judge reliability is only $\mathrm{ICC}(2, 1) = 0.56$, which is why the analysis uses the panel mean rather than one judge. Restricting to objective factual accuracy *strengthens* the tier gap and the language penalty, so the effects are not an artefact of a permissive composite. The natural next step is a blinded human-expert validation of a stratified response sample, converting an official-source-anchored result into a gold-standard-anchored one.

From an unresolved mechanism to a design that can resolve it. Neither corpus channel is established at country level, and the limitation is resolution rather than absence: three Joshi values and eleven distinct corpus-size values across 25 countries. A design that varies country coverage while holding development fixed, or a longitudinal study tracking accuracy as a country's encyclopedic footprint grows, could establish the channel that the present instrument cannot.

From a null gradient to an adequately powered test. The tier difference is established while the gradient is not, at $n = 25$ and with corrected scoring. A pre-registered replication at substantially larger n — which our open dataset makes cheap — would settle whether the gradient is genuinely absent or still underpowered.

From three native languages to the full multilingual matrix. The strongest result in the study rests on three native languages, and its mechanism is therefore reported descriptively. Native prompts were produced by a high-quality LLM translator (Claude Sonnet 4.6) rather than human translators and without a back-translation pass, so a residual translation-quality term cannot be fully excluded — a caveat that carries more weight now that H2 is the headline finding, and that back-translation would largely settle. Scaling to more languages with human-validated translation is the highest-value extension this study points to.

From one domain to a cross-domain program. All effects come from one high-stakes applied domain (air pollution policy) and one corpus family (Wikimedia); we deliberately confine the mechanistic language to this domain. The same task suite and ground-truth-registry method transfer directly to other regulated, source-anchored domains (water quality, food safety, vaccine policy).

Served-model stack as the unit of analysis. We evaluate models *as served* through fixed access stacks, which is what a public manager actually queries; a residual gap could in principle reflect the stack rather than the weights. A controlled quasi-isolation design (identical open weights across stacks) would separate the two — a clean follow-up that our per-call provenance already enables.

A tier gap that is real but not comfortable. Recomputed on the corrected scoring, the model-based corroborations are weaker than the single-judge versions reported in earlier drafts: the mixed model falls from $p = 0.007$ to $p = 0.099$, the Bayesian posterior from $P(\beta < 0) = 1.00$ to 0.984, and the E-value at the confidence limit is 1.25. We report the gap as supported because two of three approaches place the interval clear of zero and all three agree on sign and magnitude, but a reader is entitled to weigh the marginal mixed model against the bootstrap. What would settle it is not another estimator on these 25 countries but more countries, and the released pipeline makes that replication cheap.

5.8 What this study contributes

First, it is — to our knowledge — the first benchmark of LLM geographic bias in a *regulated, high-stakes applied domain* with verifiable official ground truth, moving the question beyond numeric trivia, subjective ratings, and everyday cultural knowledge to the recall and synthesis tasks a public environmental manager actually delegates. **Second**, it delivers a decisive and counterintuitive applied result: the three fixes practitioners reach for first — a local-manager persona, the local language, and a regionally tuned model — *all fail together*, the local language actively harms by 5.0 pp, and the regional model is the single worst performer of fourteen. This is the rare benchmark whose most useful contribution is telling the field what does *not* work, with the effect sizes to back it. **Third**, it demonstrates that in this literature the ground-truth key, not the sample size, is the binding constraint: the same responses yield a significant development gradient under placeholder keys and a null one under official keys, while the language penalty more than doubles. **Fourth**, it separates what a 25-country design can demonstrate (a tier difference, a language penalty, a recall floor) from what it cannot (the shape of the gradient, the corpus channel), and reports the second set as unresolved rather than as supported. **Fifth**, every reported number is recomputable from the raw data through a one-command pipeline with a built-in method-audit gate, setting a reproducibility bar — official-source ground truth, code adjudication where the answer is a number, and end-to-end auditability — that the geographic-bias literature has largely lacked.

5.9 Implications

For environmental agencies in the Global South the operational guidance is concrete: treat LLM outputs on binding standards and measured data as drafts to be checked against the official gazette; do not assume a local-language query improves accuracy, because on this evidence it degrades it; do not expect a regionally tuned small model or a role frame to close the gap. For model developers, the language penalty is the tractable lever, and it is largest exactly where the language is least represented. For the measurement community, the lesson is that a geographic-bias result can turn on the quality of the ground-truth key, so the key deserves auditing before the sample is enlarged.

6 Conclusion

Large language models are becoming working infrastructure for air pollution policy, a domain where an inaccurate answer about a binding standard, a measured concentration, or an operational response carries direct public-health stakes in the Global South, where the $PM_{2.5}$ mortality burden is heaviest [5, 42]. We measured geographic and persona-modulated bias in that domain with a 25-country, 14-model, four-language benchmark, scored against ground truth anchored to official primary sources for every country — by code wherever the answer is a register value, and by the mean of a three-vendor judge panel elsewhere.

The clearest result is that the remedies available to an agency do not work, and that the most intuitive one backfires. Asking in the country's own language *lowers* accuracy by 5.0 percentage points (Wilcoxon $p < 10^{-5}$, $n = 839$ matched cells), significantly in all three languages tested and most sharply where the language is least represented in training corpora (Hindi −12.0 pp). A local-manager persona does not narrow the gap ($p = 0.22$), and the Brazilian-Portuguese regional model is the weakest of all fourteen systems ($\delta = -0.45$). Alongside this, accuracy collapses on the recall of a binding value: 0.37 on the national standard against 0.62 for synthesis and recommendation ($\delta = -0.47$), which is the task a manager can least afford to have wrong.

A Global North/South tier gap persists (+4.6 pp, bootstrap 95% CI $[+1.2, +7.8]$, excludes zero), though modestly and with some sensitivity to specification: a Bayesian hierarchical model puts it clear of zero ($P(\beta < 0) = 0.984$) while a frequentist mixed model leaves it marginal ($p = 0.099$). Measured task by task, however, it is unambiguous and highly structured: on returning the binding national standard the Global South odds of a correct answer are 0.23 of the Global North's ($[0.19, 0.28]$), and the gap shrinks monotonically as the answer depends less on a country-specific fact, disappearing on the one task with no register value to get right. The disadvantage is concentrated rather than weak. What the gap is made of we do not resolve. The monotonic development gradient does not reach significance ($\rho = 0.36$ with HDI, $p = 0.080$; $\rho = 0.08$ at the pre-specified $n = 15$), and neither corpus channel is established at country level: the pre-specified language-corpus proxy is null ($\rho = 0.09$) and country coverage is no longer significant either ($\rho = 0.35$, $p = 0.087$; partial $p = 0.44$ after adjusting for development). We report a difference between groups that we can demonstrate and a shape across the development range that we cannot, because the design has three Joshi values and eleven distinct corpus-size values across 25 countries and lacks the resolution to separate the channels.

The message for Global South environmental agencies is concrete and largely cautionary. LLM outputs on binding standards and measured data should be treated as drafts checked against the official gazette, and none of the cheap prompt- or model-level workarounds substitutes for that verification — least of all switching to the local language, which on this evidence makes the answer worse. We read the pattern within the coloniality-of-knowledge framework [12]: the languages under-served in training corpora [10] are spoken where air pollution kills the most people, and this study turns that link from a plausible worry into a measured effect, while leaving the country-level channels unresolved.

We also report a methodological caution that we learned the hard way. Two of the five tasks were originally scored against placeholder keys, and the judge was asked to perform the arithmetic of range comparison. Rebuilding the keys from official registers and moving that comparison into code, on the same responses, cut the development gradient from $\rho = 0.51$ to $\rho = 0.36$ and more than doubled the native-language penalty. The instinct in this literature, when an effect proves fragile, is to add countries; our experience is that the ground-truth key deserves the audit first, because a defective key adds *structured* noise that concentrates exactly where official data are hardest to obtain — which is where the hypothesis lives.

Protocol, prompts, official-source ground-truth registries, analysis code, and anonymized response data are released openly, with a one-command reproduction pipeline and a method-audit gate that recomputes every reported number from the raw data. Beyond its findings, the study offers a reusable template for credible applied LLM-equity audits — official-source ground truth, code adjudication wherever the answer is a checkable number, and end-to-end reproducibility — in the high-stakes public-sector domains where these systems are now being deployed.

Data and code availability

The full protocol, prompts, and analysis code will be released on publication at github.com/Roverlucas/artigo-vies-analise-regional under MIT (code) and CC-BY-4.0 (data/prompts/documentation) licenses, together with the ground-truth registry and anonymised LLM responses. The analysis pipeline is reproducible end-to-end via `python code/run_all.py`.

Study protocol

Hypotheses, sampling frame, outcome definition, decision thresholds and multiplicity correction were fixed in a version-controlled analysis plan before confirmatory collection began. That plan was not deposited in a public registry, so readers cannot verify its timing independently of the repository history, and we do not claim pre-registration. Delivered scope also departed substantially from the planned design, which the plan itself made a condition for confirmatory status.

We therefore report the study in two layers. The first states what was specified in advance and what became of it: the three co-primary hypotheses and the secondary hypothesis are reported against their declared thresholds, including the three that failed and the one whose registered direction the data reverse. The second reports what the data suggested and the plan did not anticipate, labelled exploratory throughout, with intervals and without confirmatory language. Deviations from the plan are tabulated in the supplement. The full protocol, prompts, ground-truth registry, and analysis code will be released on publication.

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Author contributions

Following CRediT taxonomy:

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Declaration of interests

The authors declare no competing interests.

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